

$$L = 4.075 \text{ m}$$

$$\delta = 5^\circ$$

$$V_x = 30 \text{ m/s}$$

Q1

$$10 \sin\left(\frac{x}{15}\right)$$

$$\sin \quad \frac{2}{3} \cos\left(\frac{x}{15}\right)$$

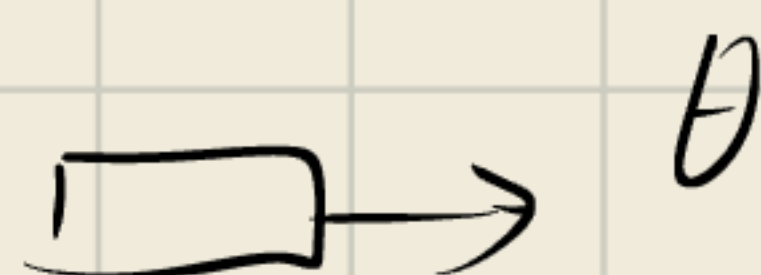
tan

$$\omega = \frac{V_x \tan \delta}{L} = \frac{d\psi}{dt}$$

$$\psi_{k+1} = \psi_k + \frac{d\psi}{dt}$$

$$X_{k+1} = X_k + \frac{dx}{dt} = X_k + V_x \cos \psi - V_y \sin \psi$$

$$Y_{k+1} = Y_k + \frac{dy}{dt} = Y_k + V_x \sin \psi + V_y \cos \psi$$



Q2

$$F(x, y) = \frac{1}{2}(x-3)^2 + 5(y-3)^2$$

$$g(x, y) = x + y - 4 \leq 0$$

$$\nabla F(x, y) = (x-3)dx + 10(y-3)dy$$

$$\mathcal{L}(x, y, \mu) = \frac{1}{2}(x-3)^2 + 5(y-3)^2 + \mu(x+y)$$

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial x} = (x-3) + \mu = 0 & \Rightarrow x \neq 0, \mu = 3-x \\ \frac{\partial \mathcal{L}}{\partial y} = 10(y-3) + \mu = 0 & \Rightarrow y \neq 0, \mu = 10(3-y) \\ \mu \geq 0 \\ \mu(x+y-4) = 0 \end{cases}$$

$$\textcircled{1} \mu > 0, (x+y-4) = 0$$

$$\begin{cases} x+y=4 \\ \mu=3-x \\ \mu=10(3-y) \\ \mu > 0 \end{cases}$$

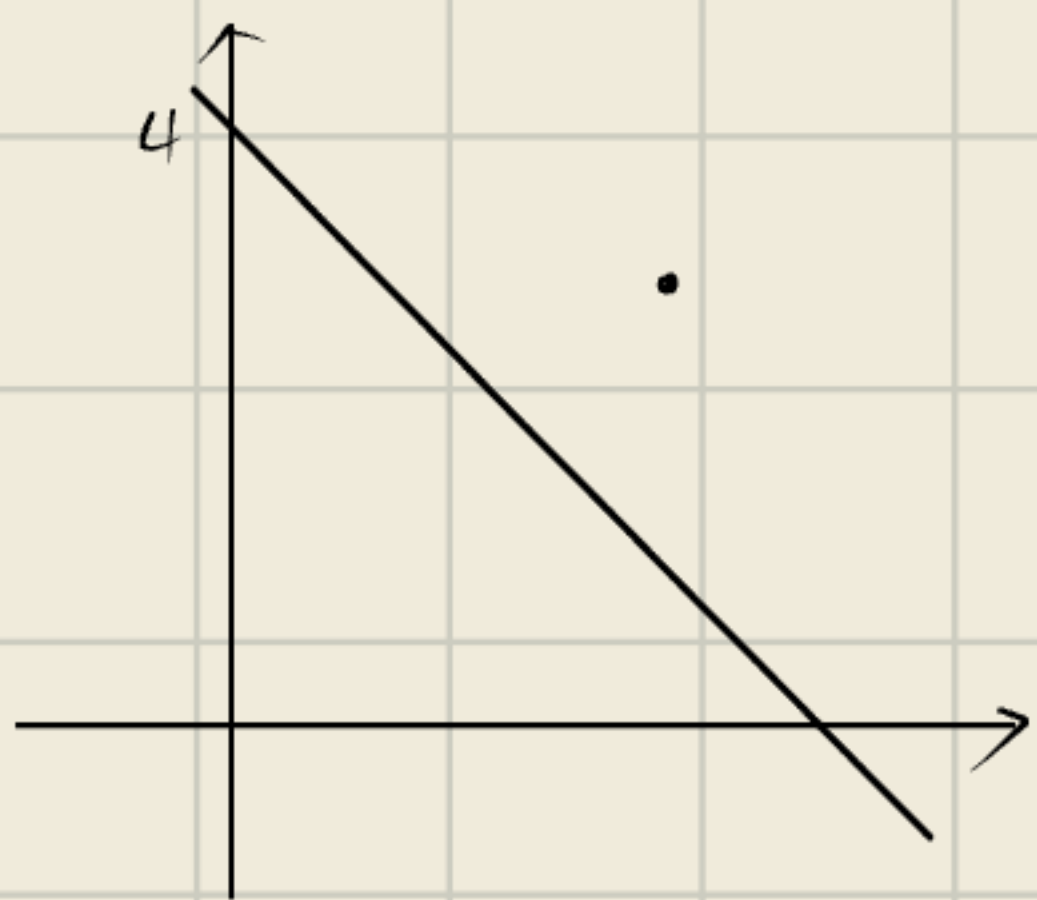
$$x = 4 - y$$

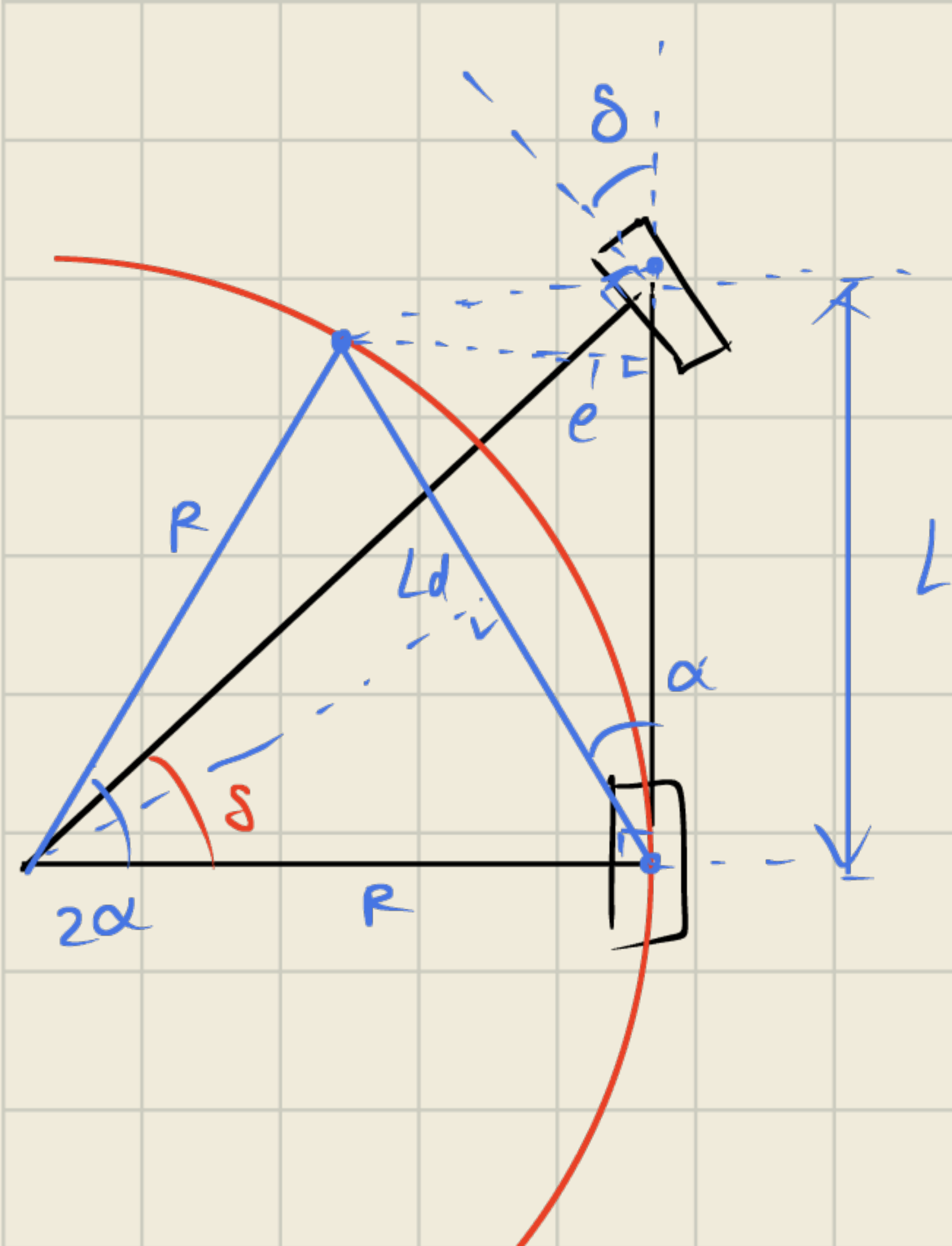
$$\Rightarrow y - 1 = 10(3 - y)$$

$$\begin{cases} y = \frac{31}{11} \\ x = \frac{13}{11} \\ \mu = \frac{20}{11} \end{cases}$$

$$\textcircled{2} \mu = 0, (x+y-4) < 0$$

$$x-3=0, y-3=0 \quad \text{舍}$$

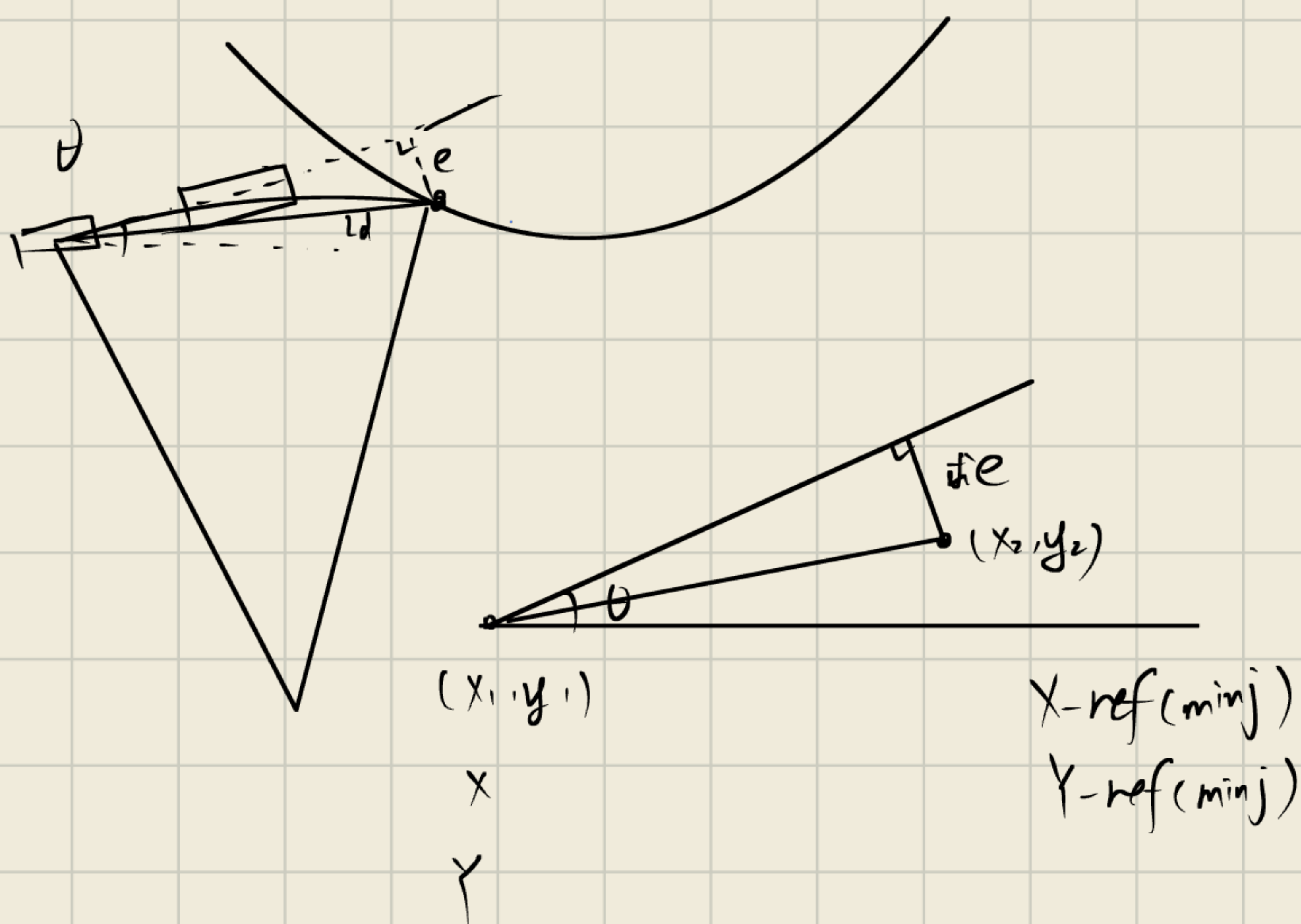




$$\begin{cases} \frac{L_d}{\sin 2\alpha} = \frac{R}{\sin(\frac{\pi}{2}-\alpha)} = \frac{R}{\cos \alpha} \\ \tan \delta = \frac{L}{R} \end{cases}$$

$$R = \frac{L_d}{2 \sin \alpha}$$

$$\delta = \arctan \frac{2L \sin \alpha}{L_d}$$



$$f(x,y) = \frac{1}{2}(x-3)^2 + 5(y-3)^2$$

$$= \frac{1}{2}x^2 - 3x + \frac{9}{2} + 5y^2 - 30y + 45$$

$$\frac{1}{2}X = [x, y] \quad f(X) = X^T \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 5 \end{bmatrix} X + X \begin{bmatrix} -3 \\ -30 \end{bmatrix}$$

$$P = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 5 \end{bmatrix} \quad Q = \begin{bmatrix} -3 \\ -30 \end{bmatrix}$$